RYA SANOVAR

(+1) 404-400-9490 ⋄ rsanovar3@gatech.edu ⋄ LinkedIn ⋄ https://rya-sanovar.github.io/

EDUCATION

Georgia Institute of Technology *Aug 2025 - Present*
Ph.D. in Computer Science
Advised by Prof. Moinuddin Qureshi, Future Architectures and Systems Lab

Birla Institute of Technology and Science, Pilani *Oct 2020 - Jun 2024*
B.E. in Electronics and Communication GPA: 8.91/10.0

WORK EXPERIENCE

Microsoft Research Cambridge, UK
Research Intern *May 2026 - Aug 2026*

- Investigated fault tolerance for disaggregated decode: Impact of network link failures and device failures on decode system throughput and tail latencies.
- Develop cost-effective scale-out network topologies for disaggregated inference resilient to network failures at datacenter scale.

Microsoft Research Bangalore, India
Research Fellow *Jan 2024 - Jul 2025*

- Developed **LeanAttention**: A hardware-aware scalable exact-attention execution mechanism that accelerates attention by 2.6x over FlashAttention-2.
  - Integrated into **ONNXRuntime**: ONNXRT LeanAttention
- Published and filed patents in key novel techniques to improve efficiency of LLM inference.

PUBLICATIONS & POSTERS

Rya Sanovar, Srikant Bharadwaj, Hritvik Taneja, Moinuddin Qureshi. *"SIFT: Selective-Index For Fast Compute of RAG Prefill by Exploiting Attention Invariance"*
Under Submission. https://arxiv.org/abs/2606.09441

Rya Sanovar, Srikant Bharadwaj, Renee St. Amant, Victor Rühle, Saravan Rajmohan. *"Lean Attention: Hardware-Aware Scalable Attention Mechanism for the Decode-Phase of Transformers"*
*The Eighth Annual Conference on Machine Learning and Systems **(MLSys)**, Santa Clara, May 2025*
https://arxiv.org/abs/2405.10480

Rya Sanovar. *"Horses for Courses: Unique Hardware Efficiency Challenges and Solutions for Prefill and Decode Inference."*
Poster presented at the *Young Professionals Symposium @ MLSys, Santa Clara, May 2025.*

PATENTS

Rya Sanovar, Srikant Bharadwaj, Victor Rühle. *"Hardware-aware attention mechanism with dynamic workload distribution for transformer models."*

RELEVANT COURSEWORK

Computer Science: Advanced Computer Architecture, Operating Systems, Machine Learning
Electrical and Electronics: Microelectronic Circuit Design, Microprocessors and Interfacing